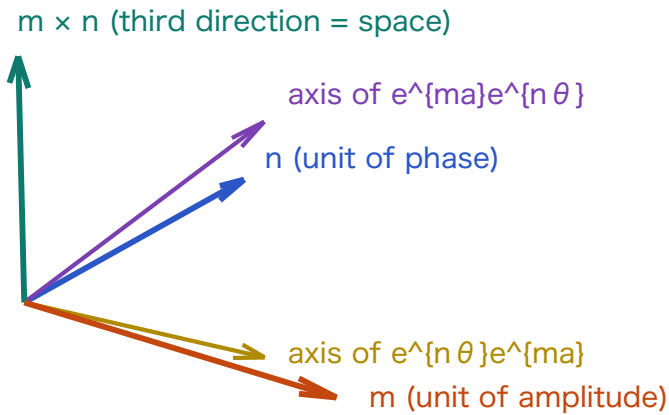
Quaternions: rotations about the amplitude axis m and the phase axis n do not commute;
the commutator e^{ma}e^{n θ} − e^{n θ}e^{ma} = 2 sin a sin θ (m × n) generates the third direction



a = π/5, θ = π/3 commutator = 1.0181 (m×n) predicted 2 sin a sin θ = 1.0181